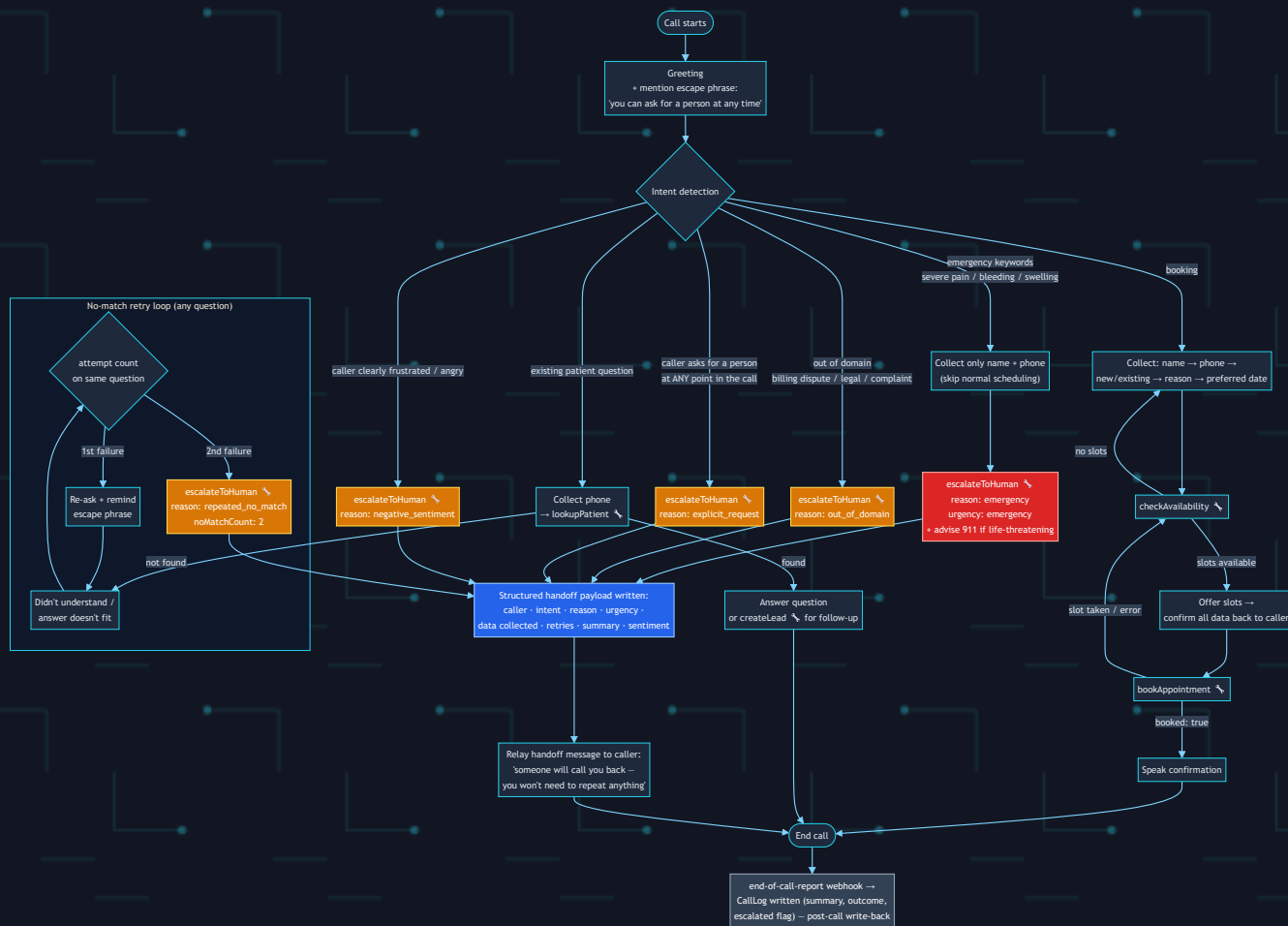


Call flow — Brightside Dental front-desk voice agent

This diagram covers the full inbound call flow, including the no-match retry loop and every escalation path. Escalation always goes through the `escalateToHuman` tool, which writes a structured handoff payload to the mock CRM (`POST /vapi/tool-calls` → `Handoff` record) before the call ends — that payload is what the human agent sees on the dashboard.



🔗 = mid-call synchronous tool call (webhook to the backend, `POST /vapi/tool-calls`).

Walkthrough: happy path (booking)

1. Caller: "Hi, I'd like to book a cleaning."
2. Agent collects name, phone, patient type, reason, preferred date — one question at a time.
3. Agent calls `checkAvailability` (never invents times), offers real open slots.

4. Caller picks a slot; agent confirms all collected data back before booking.
5. Agent calls `bookAppointment` ; on `booked: true` , speaks the confirmation and ends the call.
6. After hangup, Vapi fires `end-of-call-report` → backend writes a `CallLog` (post-call CRM write-back: summary, outcome, call id, escalated=false).

Walkthrough: escalated path (repeated no-match)

1. Caller answers the "preferred date" question with something unintelligible.
2. Agent: "Sorry, I didn't catch that — could you say the date again? You can also just say 'talk to a person' at any time." (retry 1 + escape phrase reminder)
3. Second unintelligible answer → agent does not ask a third time. It calls `escalateToHuman` with `reason: repeated_no_match` , `noMatchCount: 2` , plus everything already collected (name, phone, reason for visit) and a 1-3 sentence summary written for the human agent.
4. Backend validates and stores the structured `Handoff` ; the dashboard shows it instantly.
5. Agent tells the caller a human will call back and that they won't need to repeat anything — that promise is the entire point of the structured payload.

Design rules encoded in this flow

Rule	Where it lives
Max 2 retries per question, then escalate	System prompt — Vapi has no native retry counter, so this is prompt-enforced (see runbook, "known limitations")
Low-confidence transcripts filtered before the LLM	<code>transcriber.confidenceThreshold: 0.4</code> in <code>assistant-config.json</code>
Escape phrase always available	Mentioned at greeting + after first failed retry
Emergency skips normal flow	System prompt: collect only name+phone, escalate with <code>urgency: emergency</code>
Handoff is structured, never free-form	<code>escalateToHuman</code> JSON schema marks <code>escalationReason</code> , <code>intent</code> , <code>urgency</code> , <code>transcriptSummary</code> , <code>sentiment</code> as required; backend rejects payloads without a summary

No invented
availability /
bookings

System prompt tool rules + backend guards (slot must exist and be
unbooked)